# Pharmacy Management System

DBMS Project — Gantt Chart / Project Timeline (5 Weeks)

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## 1. Gantt Chart

| Activity | Status | W1 3-9 Nov | W2 10-16 Nov | W3 17-23 Nov | W4 24-30 Nov | W5 1-7 Dec |
|---|---|---|---|---|---|---|
| Requirements analysis & project proposal | Done | Completed | | | | |
| UI design of all 8 screens (Dashboard, Medicines, Suppliers, Customers, Purchases, Sales, Alerts, Reports) | Done | Completed | | | | |
| ER diagram & MySQL schema design (8 tables) | In Progress | | | | | |
| Schema finalization: constraints, indexes & sample data | Planned | | | | | |
| Flask project setup, MySQL connection & page templates | Planned | | | | | |
| Medicine, Category, Supplier & Customer CRUD modules | Planned | | | | | |
| Purchase module (purchase_items, automatic stock increase) | Planned | | | | | |
| Sales & Billing module (sale_items, stock decrease, bill generation) | Planned | | | | | |
| Dashboard, low-stock & expiry alerts | Planned | | | | | |
| Reports & analytics (JOIN and aggregate SQL queries) | Planned | | | | | |
| Testing, SQL query validation & debugging | Planned | | | | | |
| Documentation, demo & final presentation | Planned | | | | | |

## 2. Technologies to Be Used

| Category | Technology | Purpose |
|---|---|---|
| **Frontend** | HTML, CSS, JavaScript (Flask Jinja2 templates) | Build the responsive user interface based on the finished UI designs. |
| **Backend** | Python, Flask | Application logic, routing and database operations. |
| **Database** | MySQL (InnoDB) | Relational tables, foreign keys, transactions, JOIN and aggregate queries. |
| **DB Tools** | MySQL Workbench / phpMyAdmin | Schema design, SQL testing and data inspection. |
| **Version Control** | Git, GitHub | Track changes (repository: DBMS-Project). |

## 3. Weekly Milestones

| Week | Milestone | Expected Output |
|---|---|---|
| **W1 (3-9 Nov)** | Requirements and UI design completed | Approved proposal, README and UI screens for all 8 modules. |
| **W2 (10-16 Nov)** | Database ready, Flask connected | Final MySQL schema (8 tables) with sample data; Flask app connected to MySQL. |
| **W3 (17-23 Nov)** | Core modules working | CRUD for medicines, categories, suppliers and customers. |
| **W4 (24-30 Nov)** | Transactions and alerts working | Purchase and sales modules with automatic stock update; low-stock and expiry alerts. |
| **W5 (1-7 Dec)** | Reports, testing and presentation | Sales, purchase, low-stock and expiry reports; tested system; final demo. |

## 4. Individual Contribution

| Team Member | Assigned Responsibilities | Expected Deliverables |
|---|---|---|
| **Mahdi Hasan Shuvo** | Database design and MySQL schema; Flask backend and routes; frontend pages from the finished UI design; purchase, sales and stock logic; reports; testing and documentation. | SQL schema script, working Flask application, tested queries, project report and final demo. |